

Clinical Trials

Willem Collier, Ph.D.

Assistant Professor (USC)

Senior Statistician (Children's Oncology Group)

Email: wcollier@usc.edu

Clinical Trials Impact Our Daily Lives

- **Clinical trials support regulatory approval of new drugs:**
 - “FDA approves new non-opioid pain medication for acute pain”
 - “FDA expands approval of Wegovy to reduce cardiovascular risk”
 - “FDA approves first antibiotic in decades for uncomplicated urinary tract infections”
- **Clinical trials broadly support medical decision making, even beyond the regulatory setting:**
 - “Trial shows low-risk prostate cancers can be safely monitored without cancer-directed treatment”
 - “Trial shows safety of shortening antibiotic course”
 - “Trial shows lower dose and shorter duration of radiation therapy safe for breast cancer patients”

What is a clinical trial?

- **Clinical trial (clinicaltrials.gov):** A type of clinical study in which **participants are assigned** to groups that receive one or more **experimental interventions/treatments (or no intervention)** so that researchers can **evaluate the effects of the interventions** on health-related outcomes.
- **Randomized control trial (RCT):** Participants are RANDOMLY assigned to a treatment/intervention
 - The gold standard for evaluating efficacy and safety of a new treatment

Clinical Trials Must..

- Address a scientifically meaningful question
- Provide results that inform medical practice and, hence, benefit society
- Be ethically justifiable for trial participants
 - Minimize harm and maximize benefit
 - Control arm is selected to be the standard of care (or placebo if there are no available treatments)

Phases of a clinical trial

Phase I:

- First in-human evaluation
- Small sample (~10-60)
- GOAL: Find the “optimal” dose (balancing safety, effectiveness)
 - “What dose for phase II?”

Phase II trials:

- Larger sample (~100-200)
- GOAL: Rigorous safety assessment, preliminary efficacy
 - “Can and how to do phase III?”

Phase III trials:

- Larger sample (~200-thousands)
- GOAL: Determine the causal effect of treatment on clinical outcomes
 - “Does treatment A cause a better outcome than treatment B?”

Phase IV trials:

- Variable size
- “Post-marketing” - after FDA approval
- GOAL: Identify or monitor longer-term/rare adverse effects



Ethical Principles and Considerations for Clinical Trials

Ethical Conduct of Clinical Research:

Brief History of Failures and Corrections

- Nuremburg Doctors' Trial (1946-1947)
 - 23 Nazi physicians and administrators charged with crimes against humanity due to involuntary human experimentation. Key violations:
 - Absence of informed consent
 - Harm without therapeutic intent (harm not outweighed by benefit)
 - Vulnerable populations under coercion
- Response: Nuremburg Code (1947). Key outcomes:
 - Human participants must give voluntary, **informed**, legal consent
 - Research must address a question that WILL produce socially valuable results
 - Participants must be free to withdraw from a study at any time
 - Benefits > risks

Ethical Conduct of Clinical Research:

Brief History of Failures and Corrections (Continued)

- Post-Nuremburg unethical conduct of research continued (US)
 - Tuskegee Syphilis Study (1932-**1972**):
 - Black men in Macon County, Alabama with syphilis were observed while treatment (penicillin) was intentionally withheld despite known efficacy
 - San Antonio Contraceptive Study (late 1960s):
 - Women seeking contraceptives were given placebos without informed consent and without knowing of participation in a research study
- Response: Belmont Report (1979): Unified ethical framework for trials
 - Respect for persons:
 - Autonomy and protection for vulnerable populations
 - Beneficence:
 - Research (design, conduct, etc.) must maximize benefits and minimize harms
 - Justice:
 - Fair selection of research subjects, equitable distribution of risks and benefits

Ethical Conduct of Clinical Research: 7 Elements

1. Value
 1. Research should generate socially valuable knowledge
2. Scientific validity
 1. Methodology is rigorous with feasible design
3. Fair subject selection
 1. Selection based on scientific goals, not vulnerability or privilege
4. Favorable risk-benefit ratio
 1. Risks must be minimized and justified by potential benefits
5. Independent review
 1. Oversight by impartial body (IRB and DSMC – more next slide)
6. Informed consent/assent
 1. Voluntary, informed participation
7. Respect for enrolled subjects
 1. Protect privacy, allow withdrawal, monitor welfare

Ethical Conduct of Clinical Research: How is this ensured in practice?

- **Independent oversight: The Institutional Review Board (IRB)**
 - Primary purpose: Protect participant welfare
 - Every single trial, from proposal through completion, is overseen by one or often multiple IRB
 - Key responsibilities:
 - Review protocol and consent/assent documents for ethical acceptability
 - Evaluate risk-benefit balance
 - Ensure fair subject selection
 - Ongoing oversight (continuing review for safety, equitable participation, etc.)
 - Authority:
 - Approves, requires modifications, or disapproves research
 - Suspends or terminates studies if necessary
- NOTE: There are more review entities (e.g., DSMC)

Ethical Conduct of Clinical Research: How is this ensured in practice?

- **The informed consent process (a process, not just a document)**
- Core elements:
 - Disclosure – Potential participants given clear information RE the study
 - Comprehension – Participant understanding is ensured
 - Voluntariness – Participants must be free of coercion
 - Competence – Participants make autonomous decisions
 - Consent – Explicit, documented agreement to participate
- Key components:
 - Study purpose and duration
 - Treatment options, randomization
 - Risks and benefits
 - Right to withdraw at any time
 - Privacy protections

Clinical Trials in Pediatric Patients

- Children cannot give informed consent until they are at least 18 years old
- **Assent** is required for children under 18 years of age except in special circumstance (e.g. the child is not old enough to understand the trial and what will be asked of them)
 - Child agrees to take part in the trial
 - Several sessions may be required to help the child understand what will be asked of them
- Parent(s) must give their permission for the child to participate in the trial (with some exceptions)



**Who “starts” clinical trials? Who conducts trials?
Where do statisticians fit in?**

Trial Initiators and Trial Sponsors

Investigator Initiated Trial (IIT)



Researcher (doctor/scientist), often at academic center, seeks a trial to address a scientific question



Conducted by hospital or a “cooperative group”



Often funded by federal research grants (NIH)

Industry Sponsored Trial (IST)



Company (often pharmaceutical) seeks a trial to pursue regulatory approval of their product



Conducted by the company



Funded by the company

IIT vs IST: Examples

- IIT: A physician at USC Norris Comprehensive Cancer Center does a phase-I trial just at USC. Hospital helps to conduct. Trial is grant funded by the NCI.
- IIT: A physician at CHLA/USC initiates a phase-III trial using a cooperative group (e.g., Children's Oncology Group) because the trial is only feasible if enrolling at many hospitals. The trial is funded partly by the federal government and partly by a pharma company.
- IST: A drug company develops a new antibiotic. The drug company assembles a collection of sites that will enroll participants. The drug company funds every aspect of the trial.

Biostatisticians: Where do they work, what are their roles?

- University/academic hospital:
 - These statisticians often support IITs
 - Role: Design, monitor, analyze trial data
 - Example funding: One grant for each trial, grant provides salary support
 - Example funding: One grant for many trials under one organization (e.g., COG)
- Pharmaceutical company:
 - These statisticians support ISTs
- Contract research organization:
 - These statisticians support IITs and ISTs



Randomized phase-III trials: Why is this the “gold standard” for comparative effectiveness? Is it as easy as simply randomizing?

Statistical inference: Correlation? Causation?

- **Question for the group:**
 - Is it acceptable to give a child with leukemia a toxic chemotherapy without having a **STRONG** reason to believe it will cause a benefit (e.g., eliminate the cancer and extend time disease-free)?
- **Correlation is not a sufficient conclusion for a phase III trial.**
- **Phase III randomized trials should be designed to facilitate a causal interpretation.**

Causal inference – a thought experiment

- **Individual level causality:**
 - Imagine I am deciding between initiating two treatment options (A or B) for a disease I have.
 - I am able to create two parallel universe's where the only difference is that in universe A I get treatment A and in universe B I get treatment B.
 - If I observe a different outcome in the two universe's, what caused it?
- **Population average causality:**
 - Say we give two groups of patients from a population with a disease one of two different treatments (A or B).
 - Say these groups are, on average, ~identical with respect to every possible “risk factor” for an outcome of interest.
 - If I observe different outcome for those two groups of patients, on average, what caused the average difference?

Randomization gives us..

Physician Choice Creates Systematic Differences

Doctors (and patients) choose treatment based on patient characteristics

STUDY POPULATION



Sicker patients
more likely to get A



Healthier patients
more likely to get B

TREATMENT A



- Higher risk patients
- Worse prognosis on average

TREATMENT B



- Lower risk patients
- Better prognosis on average



Groups are systematically different

→ Outcome differences may reflect these differences,
not the treatment effect

Randomization Creates Like-to-Like Groups

Random assignment breaks systematic differences
(on average)

STUDY POPULATION



RANDOM
ASSIGNMENT

TREATMENT A



- Similar mix of patients
- Similar average risk

TREATMENT B



- Similar mix of patients
- Similar average risk



Groups are like-to-like (on average)

→ Outcome differences can be attributed
to the treatment

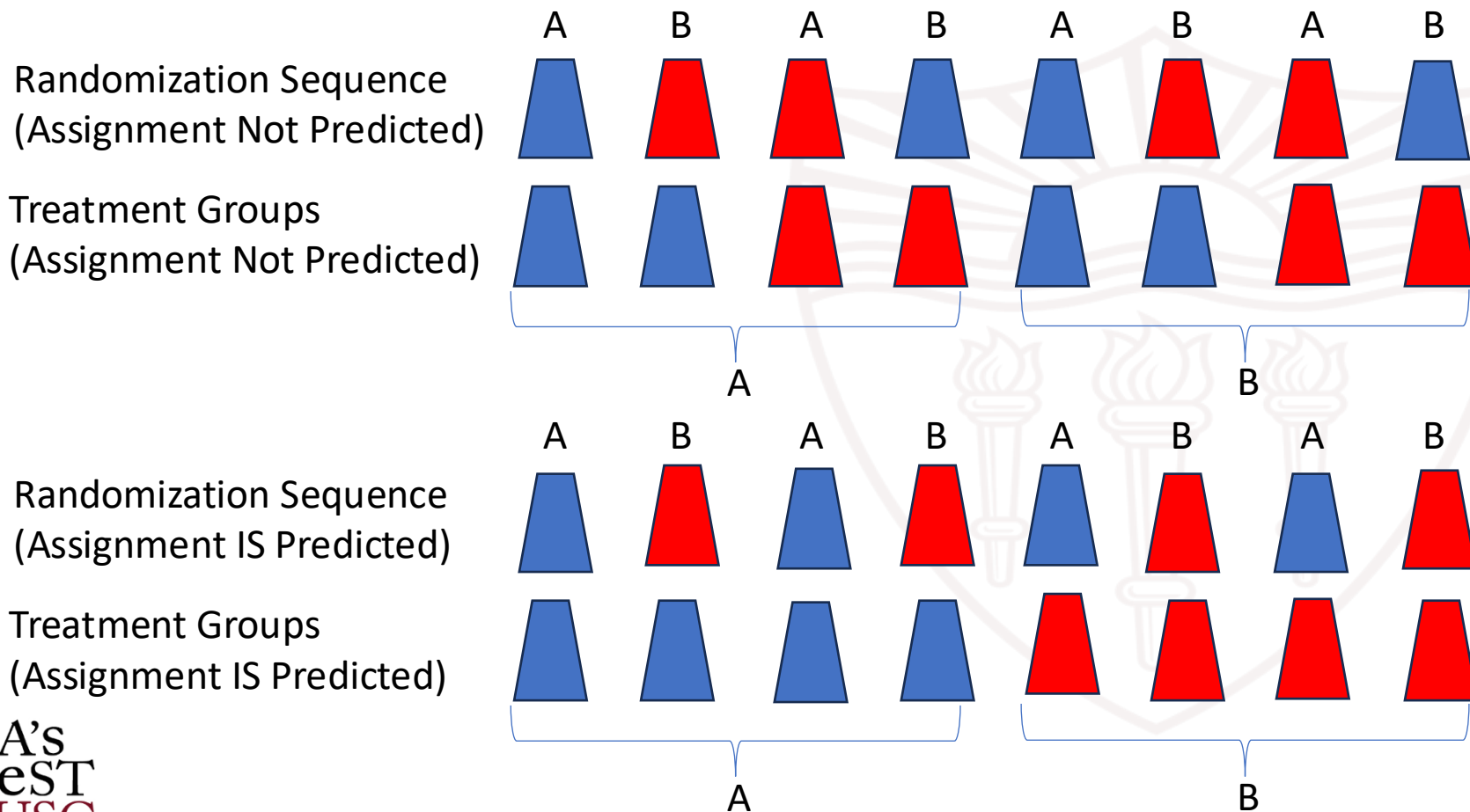
Randomization Continued

- Randomization helps to ensure that treatment assignment is not systematically determined by patient characteristics.
- This helps to ensure randomized groups differ, on average, only in their treatment assignment.
- Under the right conditions, differences in outcomes are attributable to the treatment assignment, and not “other reasons”: causation.

What breaks randomization? When things go wrong..

Example Problem	What can be done?
Physicians predict randomization sequence, enroll accordingly -> selection bias/confounding	• Randomization methods (make unpredictable) • Blinding (physicians, patients) with placebo control
Patients analyzed are not as randomized: Example 1: Post-randomization exclusion criteria	• Design to minimize post-randomization exclusion • Stratified/adjusted analyses • Sensitivity analyses
Patients analyzed are not as randomized: Example 2: Missing data	• Statistical analysis plan for handling missing data • Assumptions clearly outlined • Sensitivity analyses

Breaking Randomization – Example 1: Sequence Prediction



Breaking Randomization – Example 1 “Solutions”

- Blinding (e.g., investigators (physicians) and/or patients do not know whether they received treatment A or treatment B (placebo controlled)).
- Randomization methods (e.g., randomization happens within “blocks” which have random, variable size).
- Multicenter trial (physician doesn’t know when and how last patient was randomized because it occurred at a different site, meaning they cannot predict the next treatment assignment at their center)
- Key: The statistician plays a role in this!

Breaking Randomization – Example 2: Missing Data

	Arm A	Arm B
Proportion with “Risk Factor” in Overall Cohort Enrolled	50%	50%
High-risk patients more likely to withdraw from the study, which is impacted by treatment	10% with “Risk Factor” withdraw early (missing “end of study” data)	90% with “Risk Factor” withdraw early (missing “end of study” data)
Proportion with “Risk Factor” in Cohort with Complete Data	~47%	~9%
Problem	The two groups of patients with complete data are not alike	

Breaking Randomization – Example 2 “Solutions”

- Trial design
 - Convenience and support for participants
 - Choose follow-up schedules aligned with routine clinical care and which are practical for completion
- Monitoring
 - Monitor missingness, identify if patients missing differ from those who are completing evaluations
 - Can update protocol accordingly (sometimes)
- Analysis
 - Methods which utilize intermediate data for patients with some missing data
 - Missing data handling methods
 - *Sensitivity analyses (“how much do my primary results change based on the assumptions I make regarding missing data?”)

Randomization Summary

- Randomization is a fundamental component of drawing strong, causal conclusions in comparative clinical trials
- Randomization helps to ensure randomized treatment groups are similar with respect to possible risk factors for the outcome
 - Hence, outcome differences are attributable to the treatments assigned
- The presence of randomized treatment assignment alone does NOT guarantee that the trial supports strong conclusions
 - Randomization can be “broken” for many reasons
- The statistician plays a key role in aligning the design, monitoring, and analysis of a trial to support strong, causal trial interpretation

The Statistician's Job:
Design
Monitoring
Analysis and Results Dissemination

Designing clinical trials

- Designing a clinical trial is a highly collaborative process
 - Involves subject matter experts (e.g., physicians, psychologists), statisticians, pharmacists, patient advocates, regulators, etc.
- The statistician aids in choosing:
 - Study population (inclusion/exclusion criteria)
 - Outcomes and endpoints (primary, secondary, exploratory)
 - Target measure of treatment effect on each endpoint (e.g., risk ratio)
 - Statistical analysis plan
 - Hypotheses tested
 - Statistical method (testing procedure, models, estimation and inferential methods)
 - Plans for handling “intercurrent events”
 - Sensitivity analyses
 - Strategies for presenting results of analyses
 - Determining type-I error targeted and controlled
 - Powering the trial (determining the sample size needed)

Designing clinical trials - Example

Physicians and psychologists want to compare two treatments for neurocognitive protection for patients who are receiving cranial radiation therapy (CRT)

Statistician and domain experts come to an agreement on:

- Cognition to be measured using a “3-domain Z-score” (outcome)
- Cognition measured up-to 1-year after CRT start (endpoint), obtained monthly during 1-year of observation time
- Target summary measure of treatment effect: The expected difference in Z-scores at 1-year across treatment groups among initiators of treatment

Statistician designs the analysis plan. Example decisions:

- The 3-domain Z-score will be modeled as a function of arm and time using a longitudinal marginal model
 - Regression coefficients estimated under the model will be used for a hypothesis test
- All patients who initiate treatment will be included in model fitting, the statistician articulates assumptions made
- The statistician determines 300 patients (150 per-arm) are needed to “power” the trial at 80% under a pre-specified hypothesized effect size and “significance level”

Monitoring Clinical Trials

- Once a clinical trial is “activated” (is approved by various review and regulatory entities), the data must be monitored for many reasons:
 - Is accrual proceeding as expected?
 - Example question: If not, should the trial stop due to infeasibility?
 - Can the trial operate as intended?
 - Example concern: Too many patients are skipping study assessments because they are overly burdensome.
 - Is the study safe?
 - Example question: Are more participants experiencing toxicities than we would expect them to if not for enrollment onto the trial?
 - Under planned analyses: Does the trial’s data, at pre-planned analysis timepoints, provide early evidence that the trial should stop?
 - Example: At half of the planned sample size, the data indicates the treatment is not sufficiently beneficial to warrant accrual to the full sample size.

Monitoring Clinical Trials - Notes

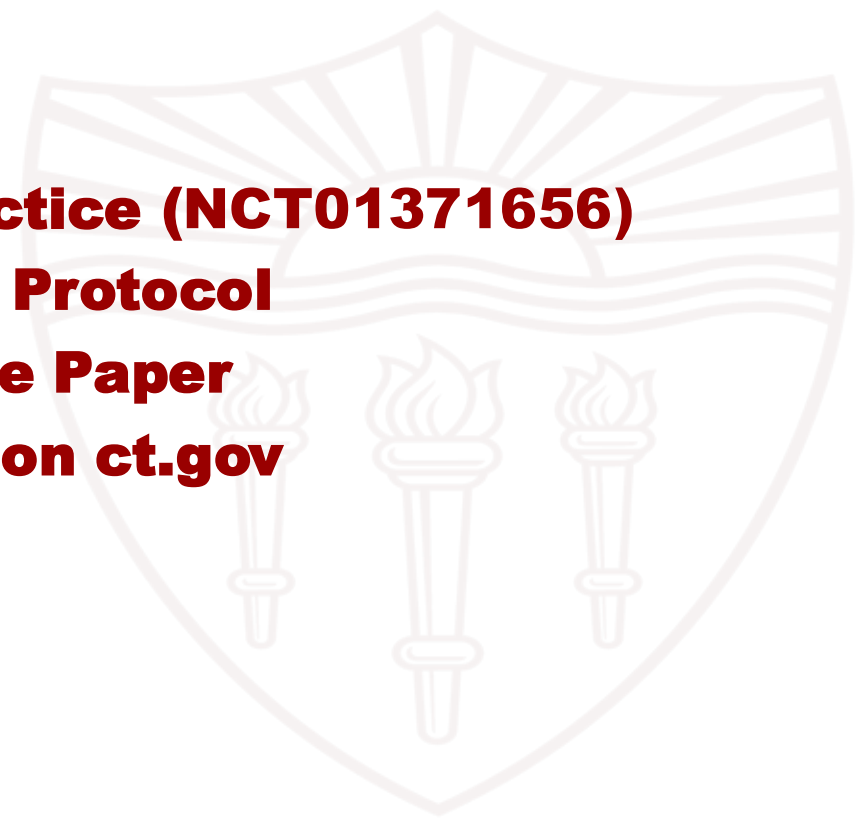
- Statisticians will prepare reports for monitoring of trials for:
 - Data Safety and Monitoring Committee (DSMC) (annual, biannual, post-hoc)
 - The study committee (monthly meetings)
 - IRB (e.g., the Pediatric Central IRB for COG trials – annual reports)
- Safety is usually evaluated in two ways:
 - Planned analyses of toxicities of concern (involves planned statistical analysis)
 - Unplanned assessments (analyses are not planned, but can arise for unanticipated concerns)
- Analyzing the primary outcome:
 - The primary outcome should only be reported on, by arm, at the end of the study except under pre-planned “interim analyses”
 - We cannot attempt to learn about or release the results of the trial before it is completed

Analyzing Clinical Trials

- At the end of the trial, primary, secondary, and exploratory outcomes are analyzed, to the extent possible, according to pre-planned protocolized statistical plans.
- Results are reported via:
 - Clinicaltrials.gov (government hosted website used to facilitate transparency in research findings for the general public)
 - Conference presentations
 - Peer-reviewed journal articles
- The statistician:
 - Manages data and implements analyses
 - Determines how results should be presented (tables, figures, which metrics to report)
 - Writes!

Analyzing Clinical Trials – Core Elements of a Paper

- Describe who was enrolled
 - Demographic and clinical characteristics of the study cohort
 - Helps to inform readers about the applicability of the trial's findings
- Describe how patients progressed through the study
 - How many patients didn't complete planned treatments?
 - How many patients withdrew from the study early?
- Results of the primary analysis
 - Do the data indicate one treatment relative to another was as beneficial as hypothesized with respect to the clinically most important outcome?
- Results of secondary and exploratory outcome analyses
 - Typically used to contextualize the results of the primary outcome
 - In many cases, cannot be used to decide whether or not a new treatment is “efficacious”, “non-inferior”, etc.



Clinical Trials in Practice (NCT01371656)

Example Protocol

Example Paper

Example on [ct.gov](https://clinicaltrials.gov)